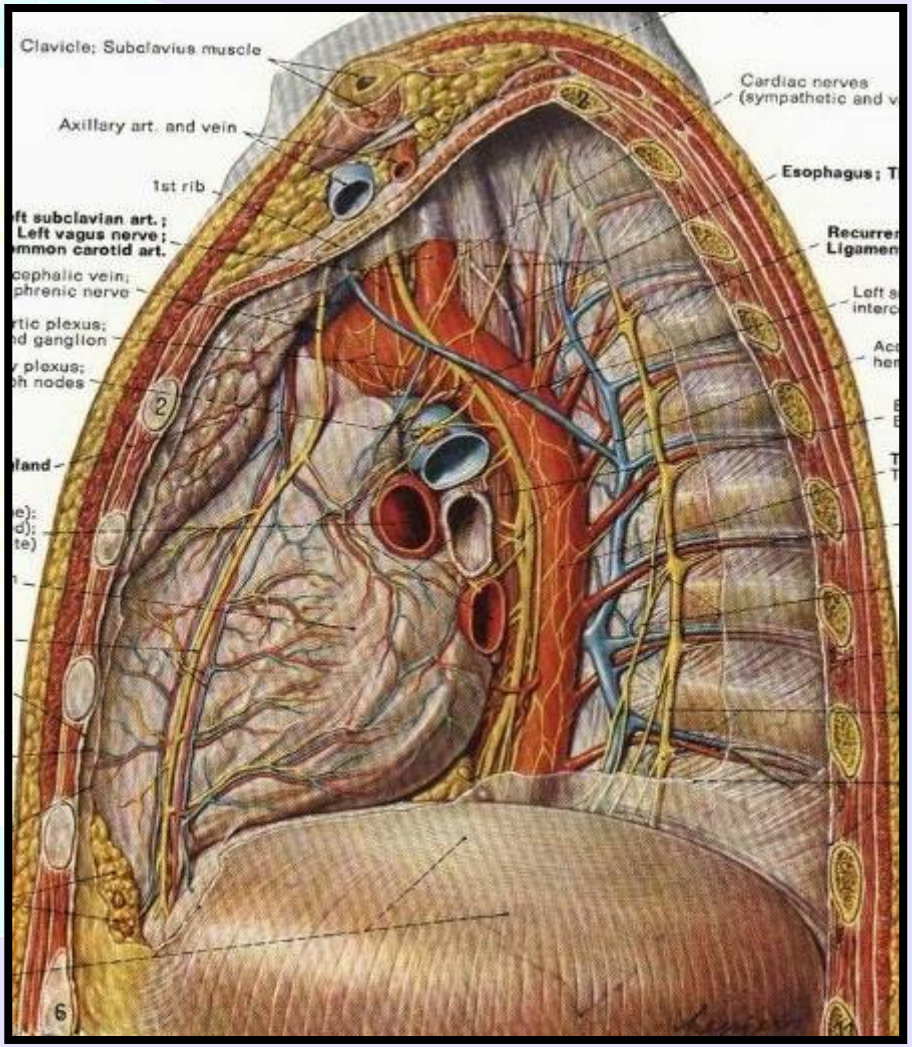
ANTERIOR AND POSTERIOR MEDIASTINUM

LEARNING OBJECTIVES

- Identify the space between the two pleural cavities.
- Know the boundaries and subdivisions of mediastinum.
- Know the contents and the relationships among structures of Anterior and Posterior mediastina.

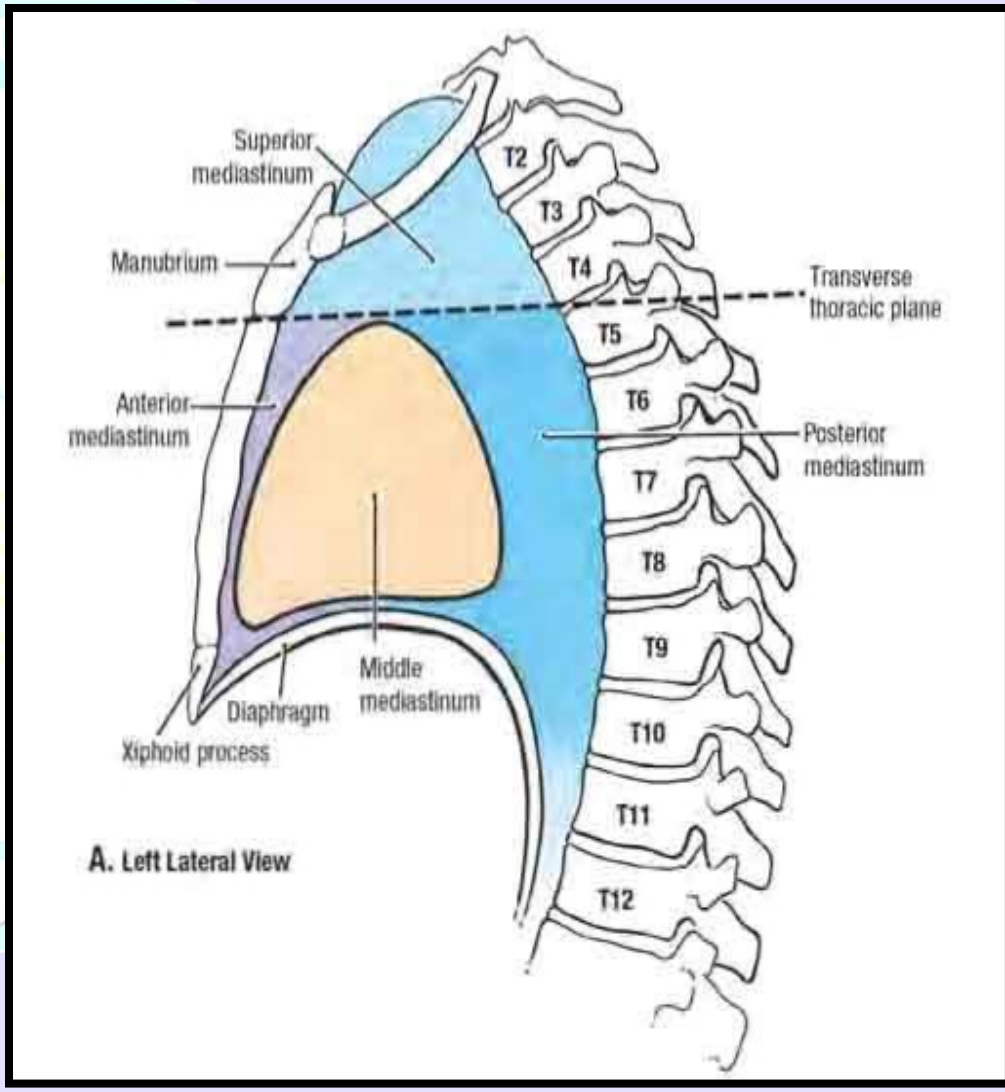
MEDIASTINUM

- The space between the pleural cavities occupying the center of thoracic cavity is the mediastinum
- extends superiorly to the thoracic inlet and root of neck.
- Inferiorly extends to the diaphragm.



BOUNDARIES OF MEDIASTINUM

- **Anteriorly**
  sternum
- **Posteriorly**
  vertebral column
- **Superiorly:**
  thoracic inlet
- **Inferiorly**
  diaphragm
- **On each side:**
  lungs & pleura

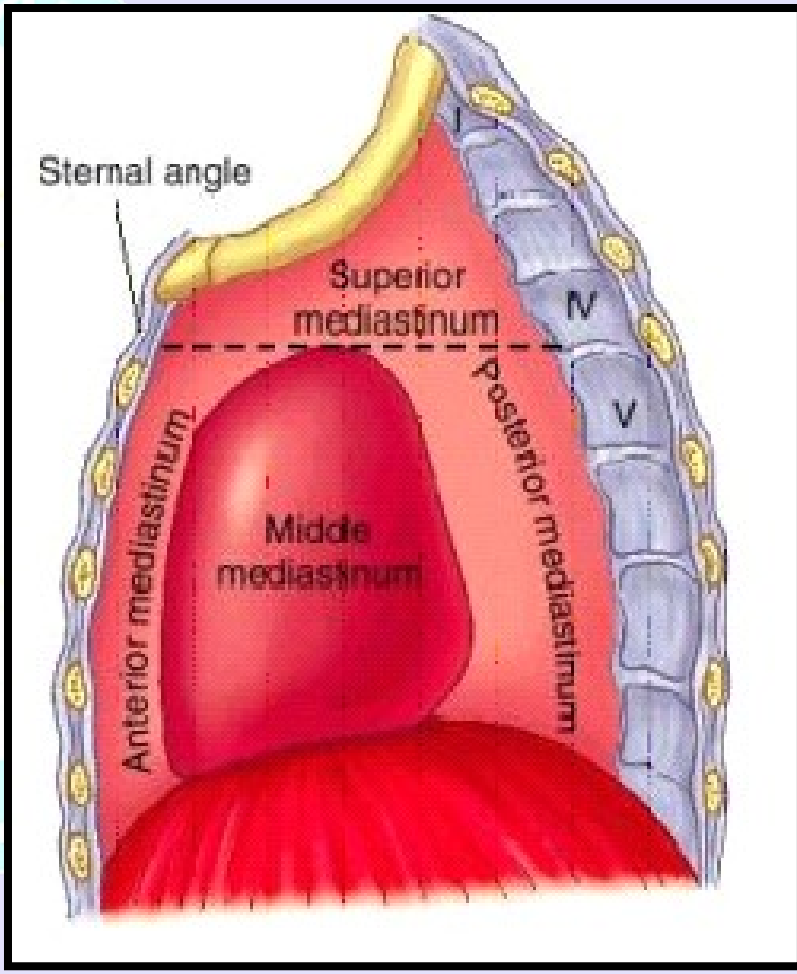


DIVISION OF MEDIASTINUM

Mediastinum is divided into

- **superior** and
- **inferior mediastinum**

by imaginary line passing anteriorly through sternal angle to the lower border of body of fourth thoracic vertebra posteriorly.



INFERIOR MEDIASTINUM

divided into three parts by the presence of heart and pericardium

- **ANTERIOR MEDIASTINUM**

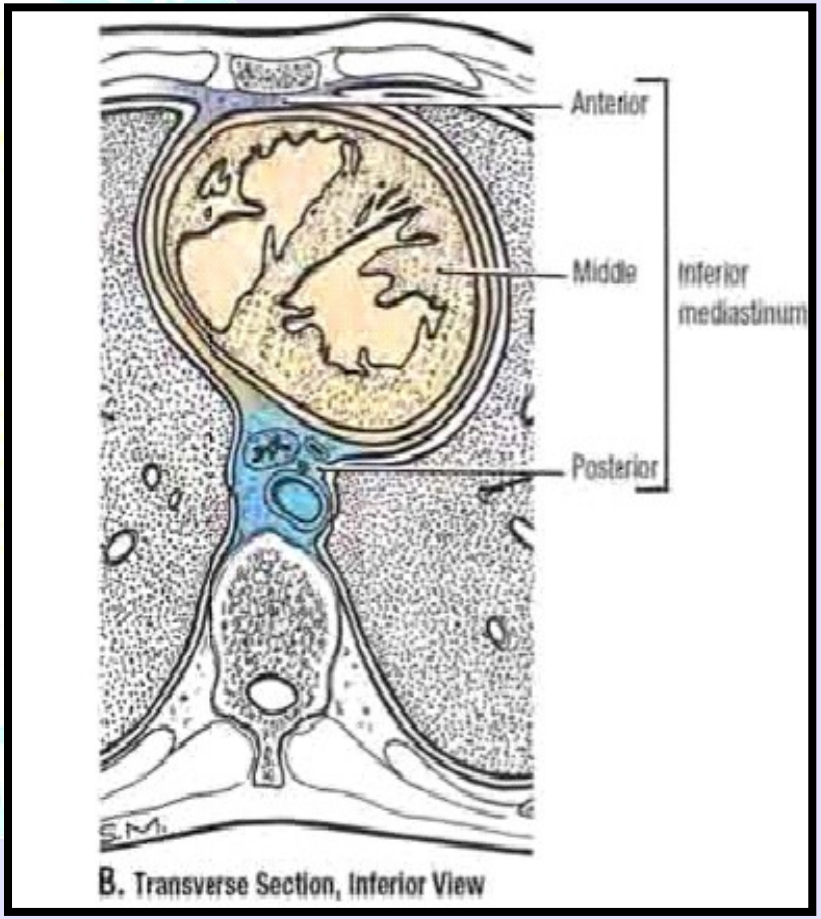
Lies In front Of The Pericardium

- **MIDDLE MEDIASTINUM**

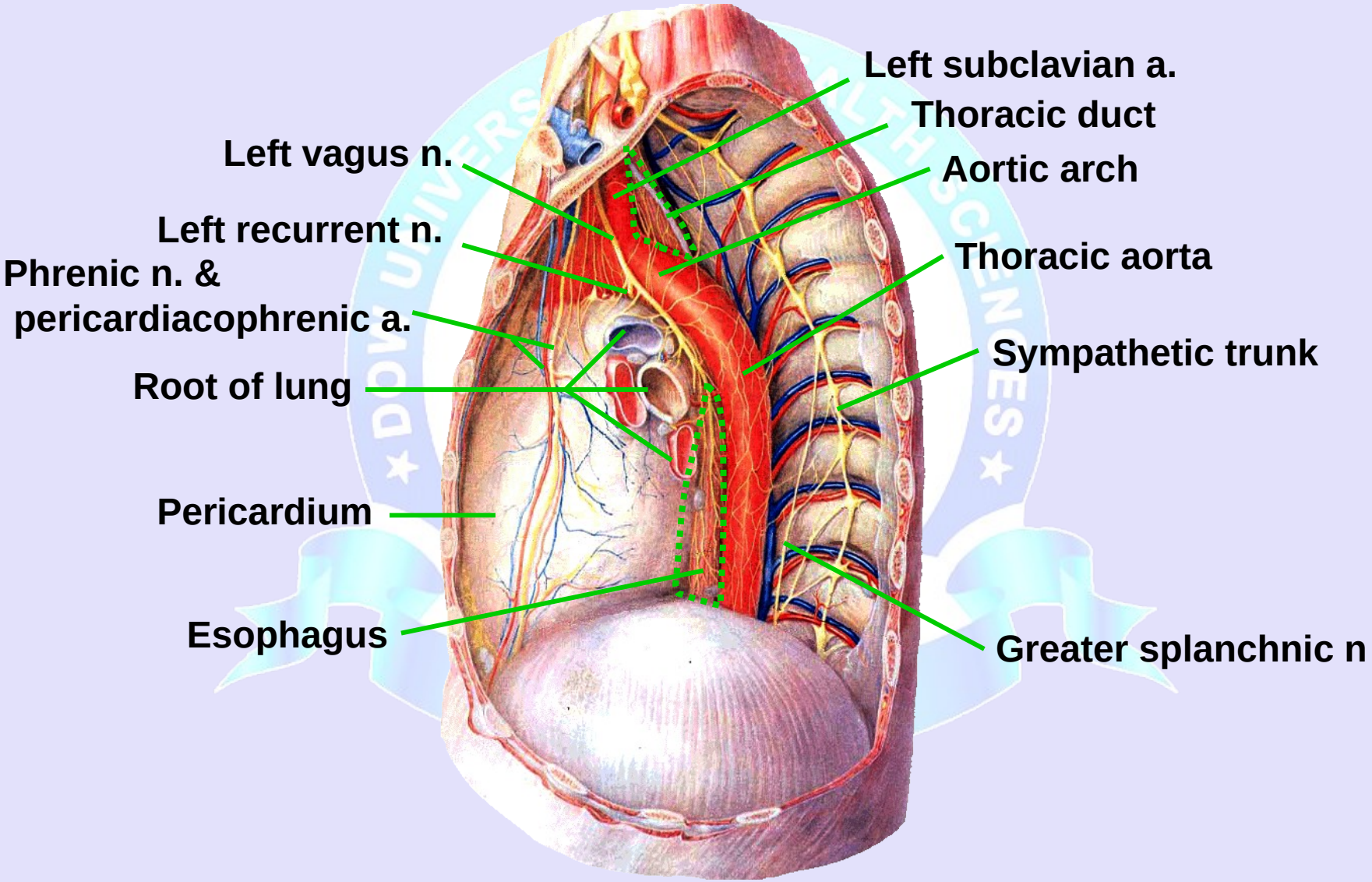
Contains Heart And Pericardium

- **POSTERIOR MEDIASTINUM**

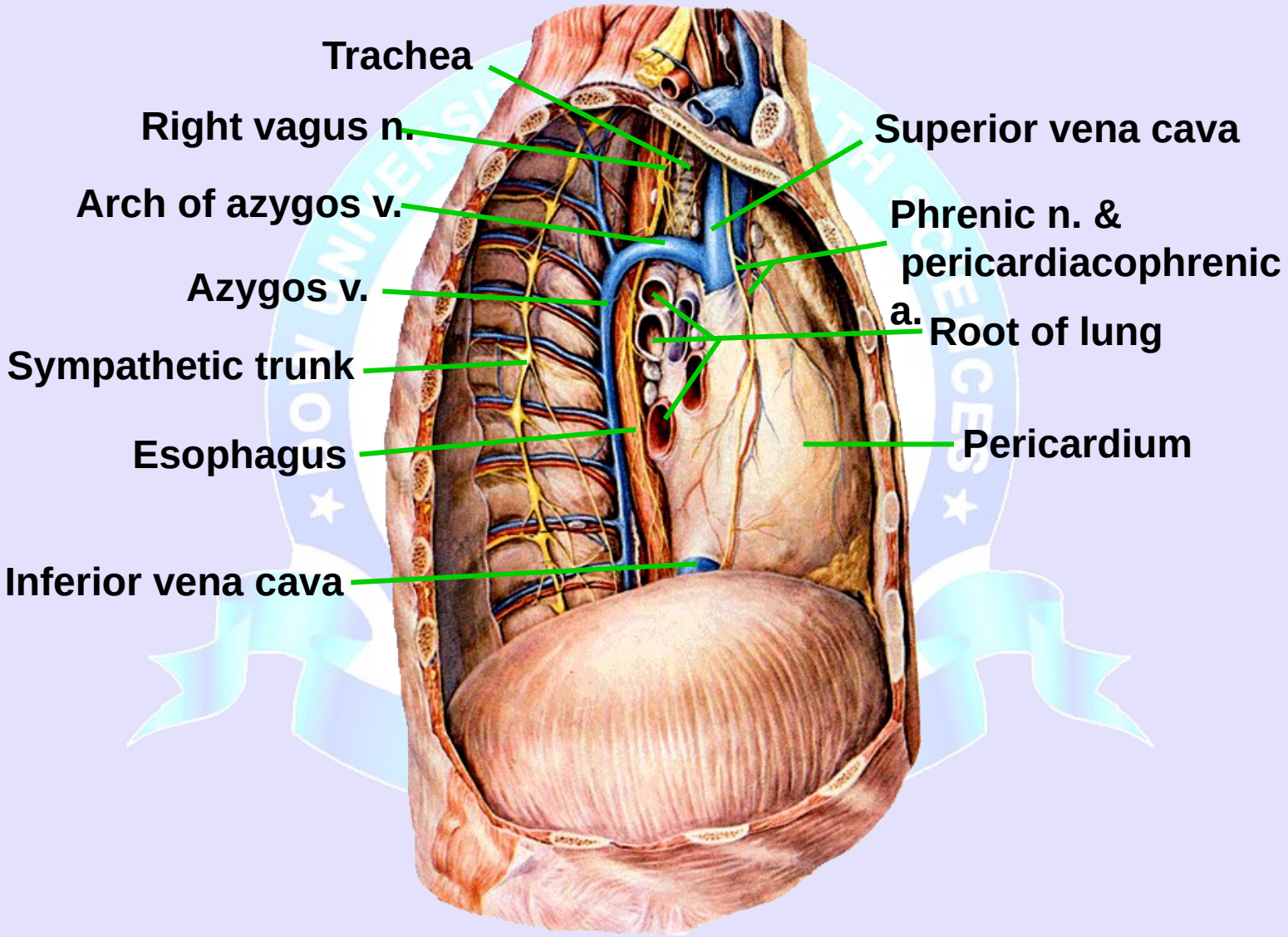
Lies Behind The Pericardium



LEFT SIDE OF MEDIASTINUM

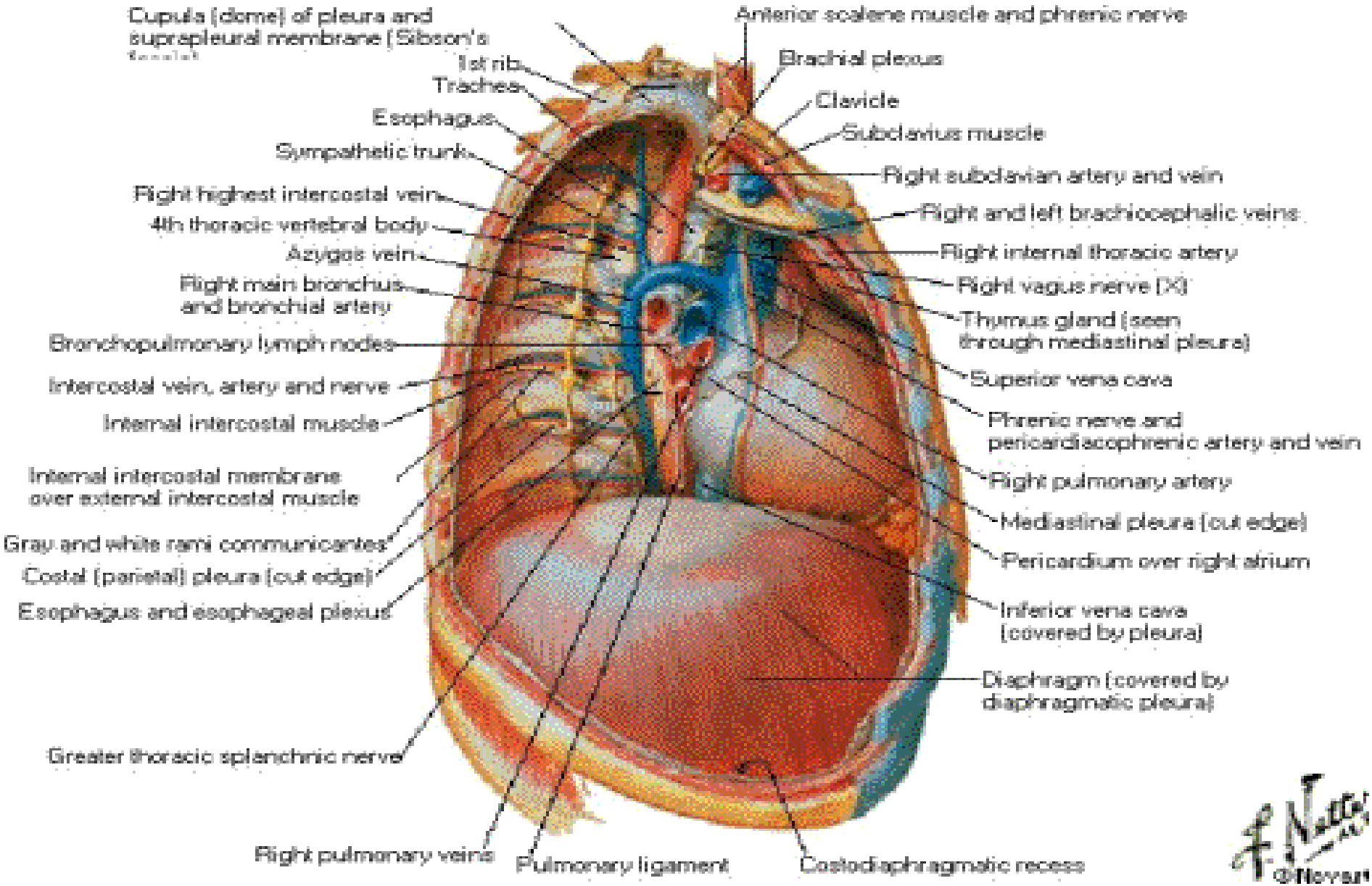


RIGHT SIDE OF MEDIASTINUM



Mediastinum

Right Lateral View



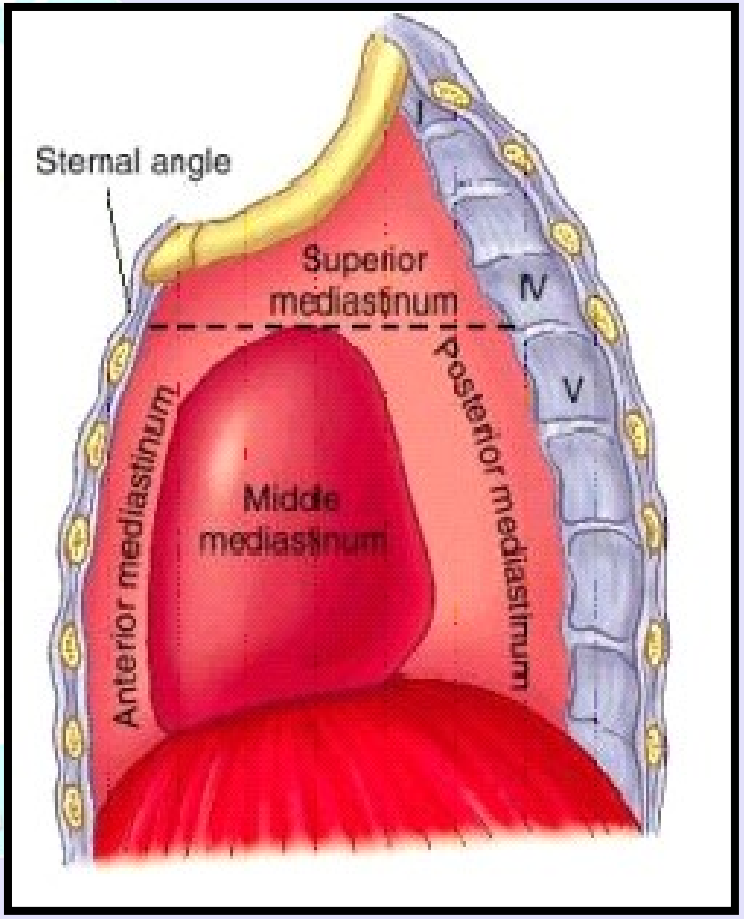
INFERIOR MEDIASTINUM

ANTERIOR MEDIASTINUM

- **Location**

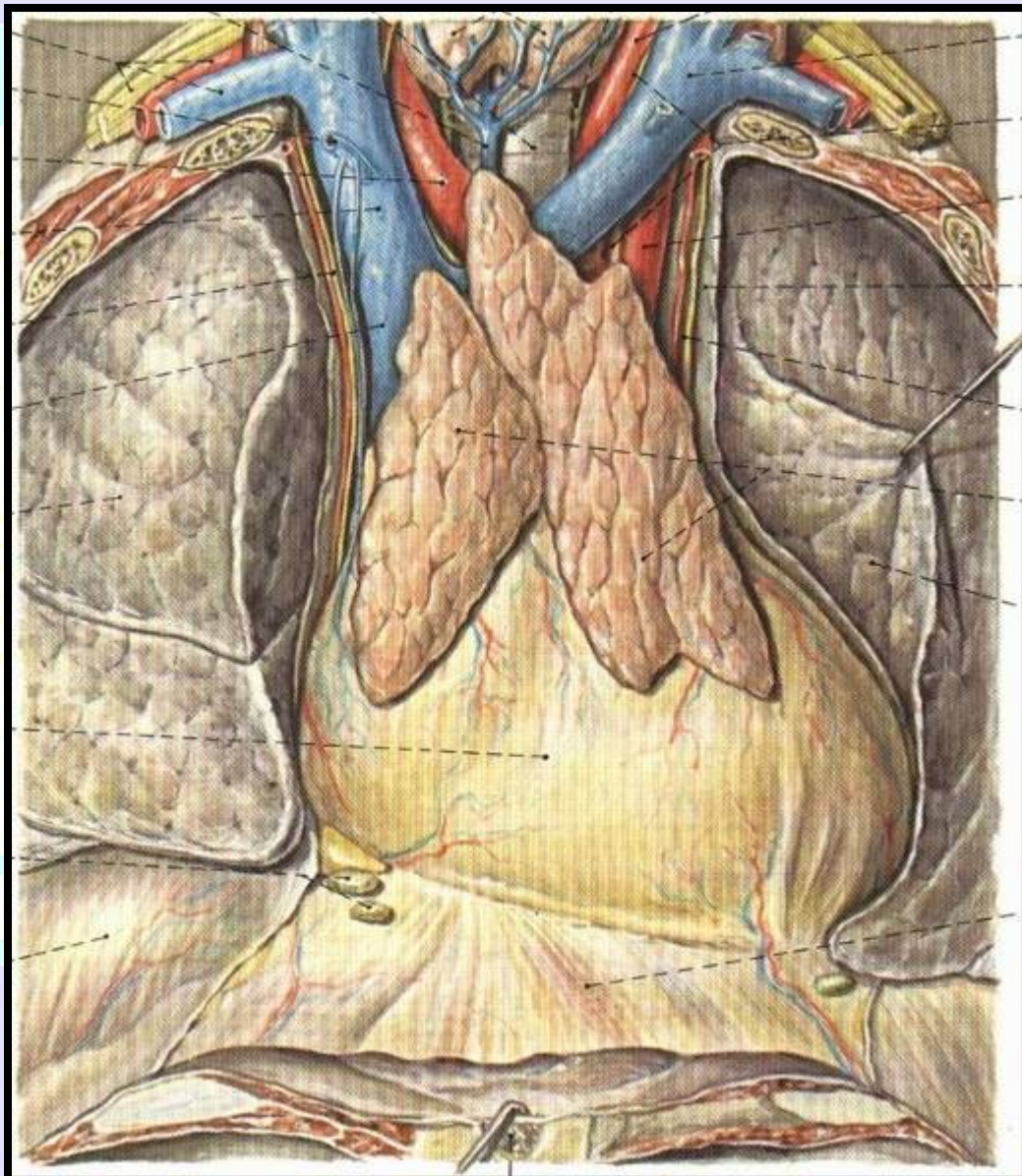
  posterior to body of sternum and attached costal cartilages, anterior to heart and pericardium

- **Contents**
  - **Fat**
  - **Remnants of thymus gland**
  - **Anterior mediastinal lymph nodes**



THYMUS

The remains of the thymus embedded in fatty connective tissue immediately posterior to manubrium sterni.



THYMUS

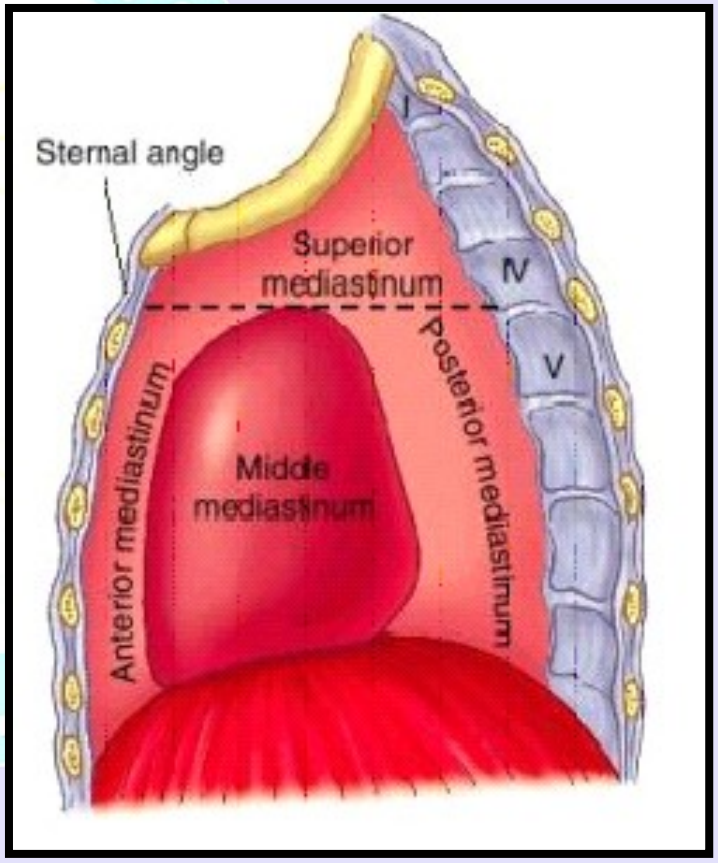
- The organ reaches its greatest size relative to the remainder of the body at birth.
- It continues to enlarge until puberty, when it gradually starts to atrophy, and very little is present in old age.
- It receives its arterial supply from the internal thoracic arteries.

INFERIOR MEDIASTINUM

POSTERIOR MEDIASTINUM

Location:

- **posterior to heart and pericardium,**
- **anterior to vertebrae T5-T12**

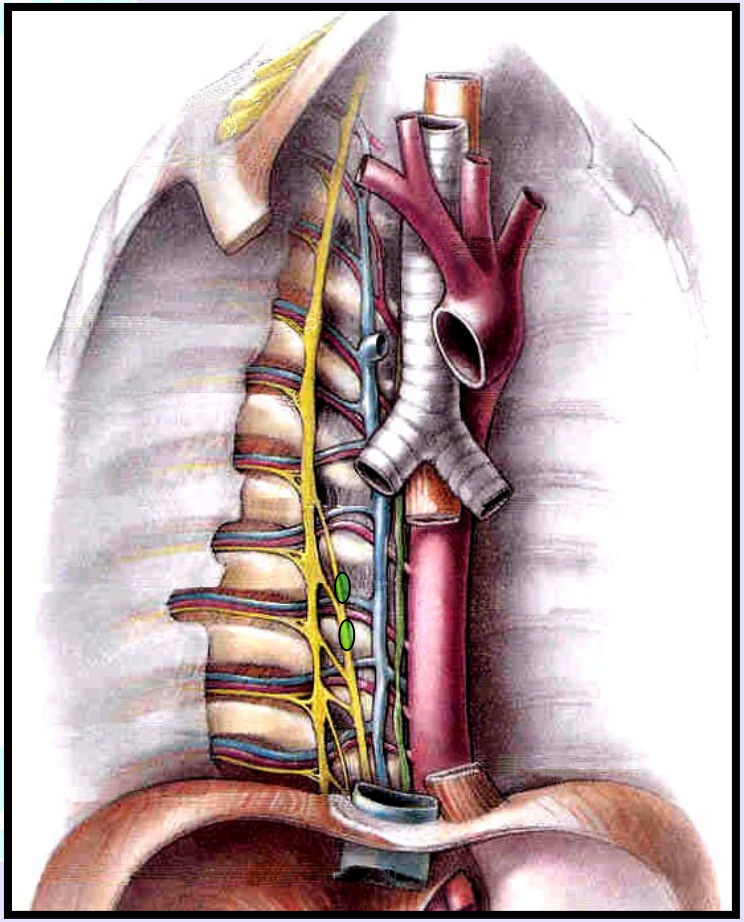


INFERIOR MEDIASTINUM

POSTERIOR MEDIASTINUM

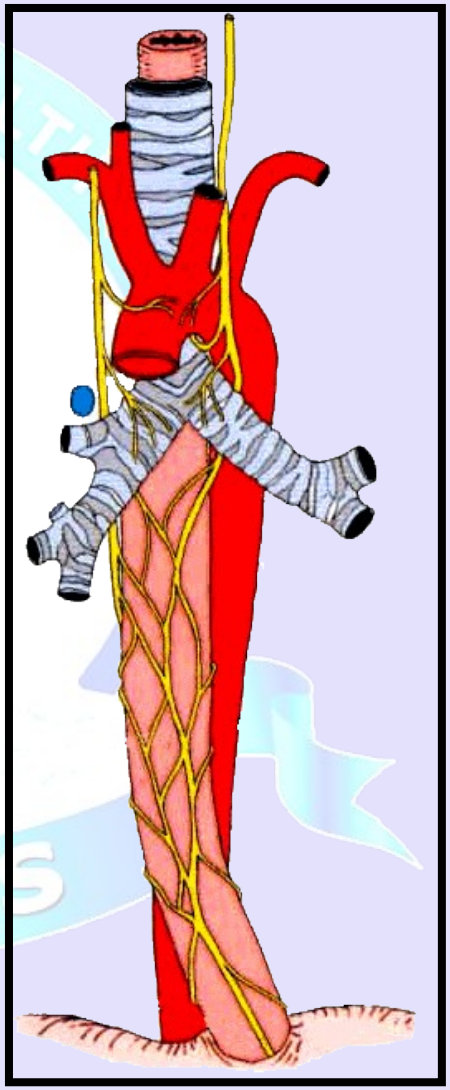
Contents:

- Esophagus
- Vagi and sphlanchnic nerves
- Thoracic aorta and its branches
- Azygos system of veins
- Thoracic duct
- Thoracic sympathetic trunk
- Posterior mediastinal lymph nodes



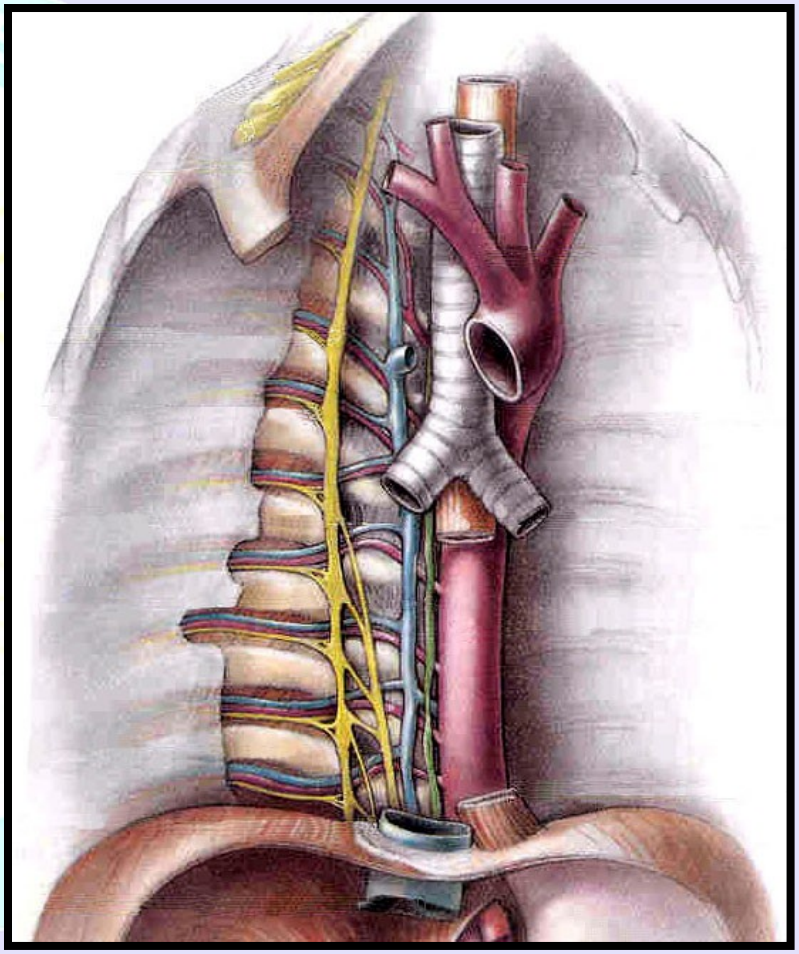
RELATIONS OF ESOPHAGUS

- **Anteriorly**
  - Trachea
  - Bifurcation of trachea
  - Left principal bronchus
  - Left recurrent n.
  - Right pulmonary a.
  - Anterior esophageal plexus
  - Pericardium
  - Left atrium
  - Diaphragm



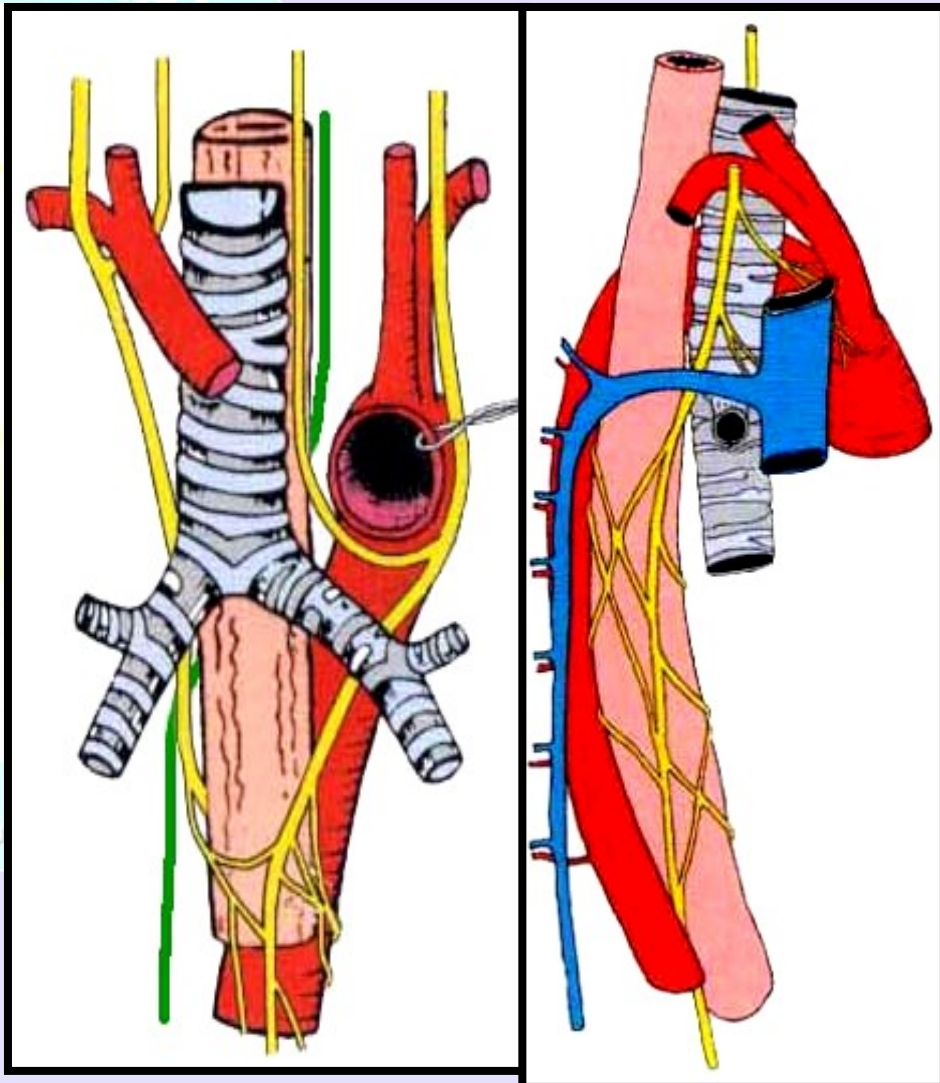
RELATIONS OF ESOPHAGUS

- **Posteriorly**
  - Posterior esophageal plexus
  - Thoracic aorta
  - Thoracic duct
  - Azygos v.
  - Hemiazygos v.
  - Accessory hemiazygos v.
  - Right posterior intercostal v.



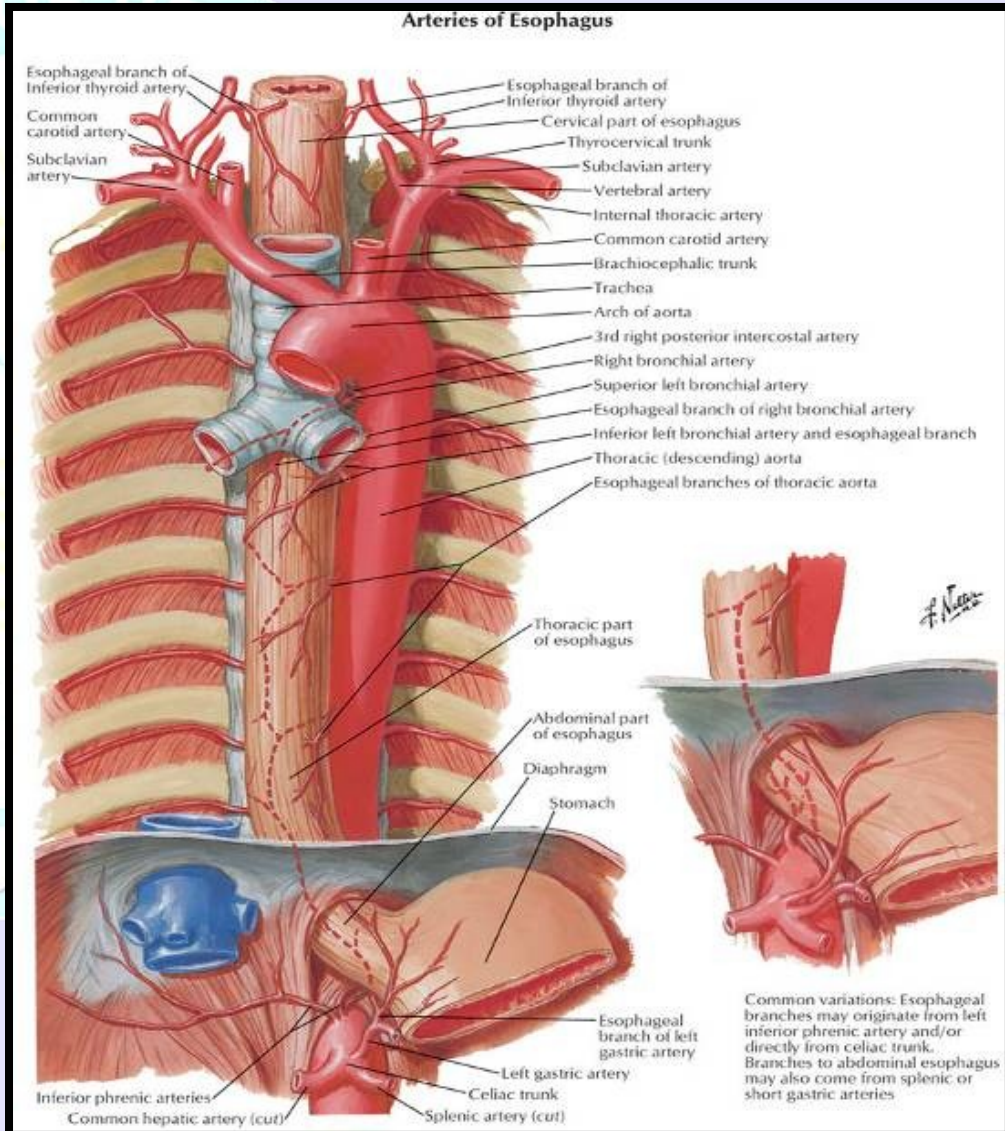
RELATIONS OF ESOPHAGUS

- **Left**
  - Left common carotid a.
  - Left subclavian a.
  - Aortic arch
  - Thoracic aorta
  - Superior part of thoracic duct
- **Right**
  - Arch of azygos v.



BLOOD SUPPLY OF ESOPHAGUS

- **Bronchial artery.**
- **Thoracic aorta.**
- **Left gastric artery.**
- **Left inferior phrenic artery.**



INNERVATION OF ESOPHAGUS

Esophageal plexus:

- Continuation of posterior pulmonary plexus.
- Formed by right and left vagus nerves:
- Right vagus nerve → posterior vagus nerve.
- Left vagus nerve → anterior vagus nerve.

INNERVATION OF ESOPHAGUS

Upper third:

Voluntary muscle.

Innervated by recurrent laryngeal nerve.

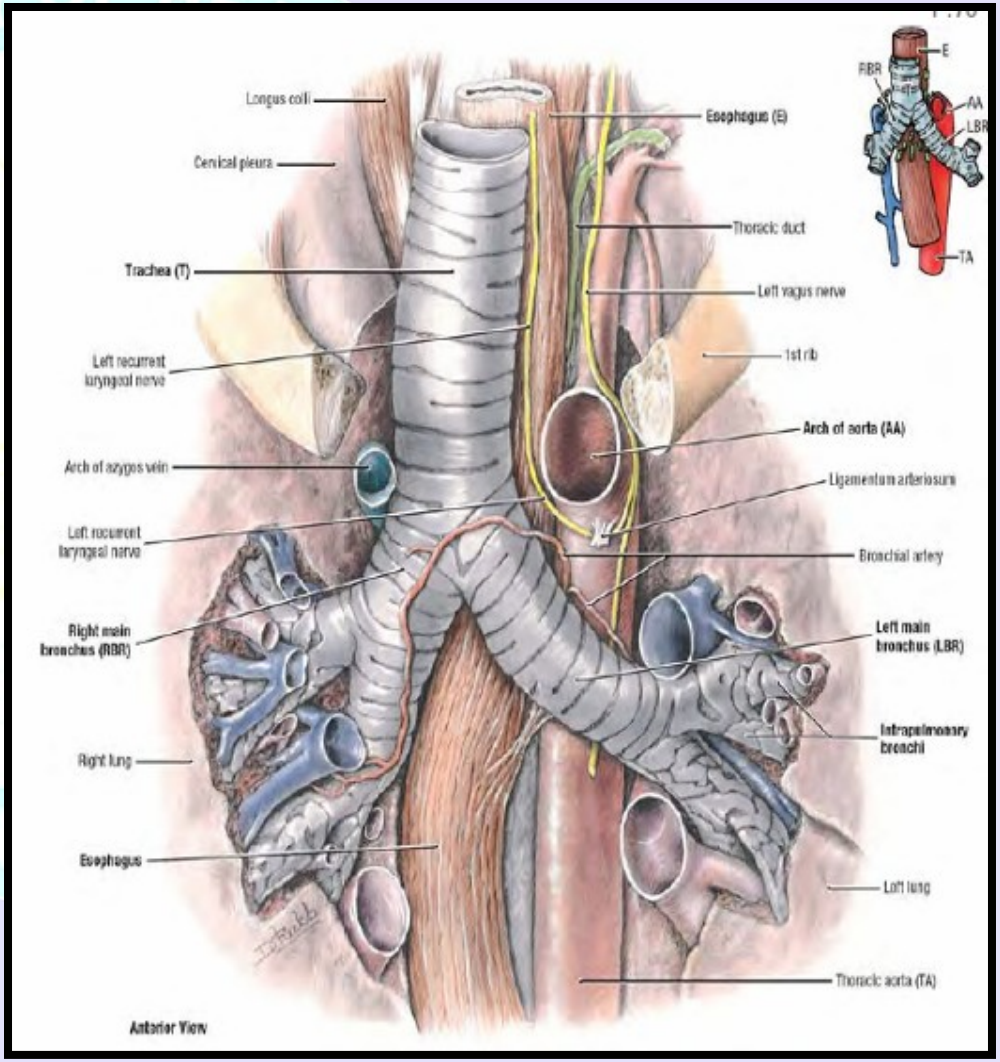
Lower two-thirds:

Involuntary muscle.

Innervated by vagus and sympathetic chain.

DESCENDING THORACIC AORTA

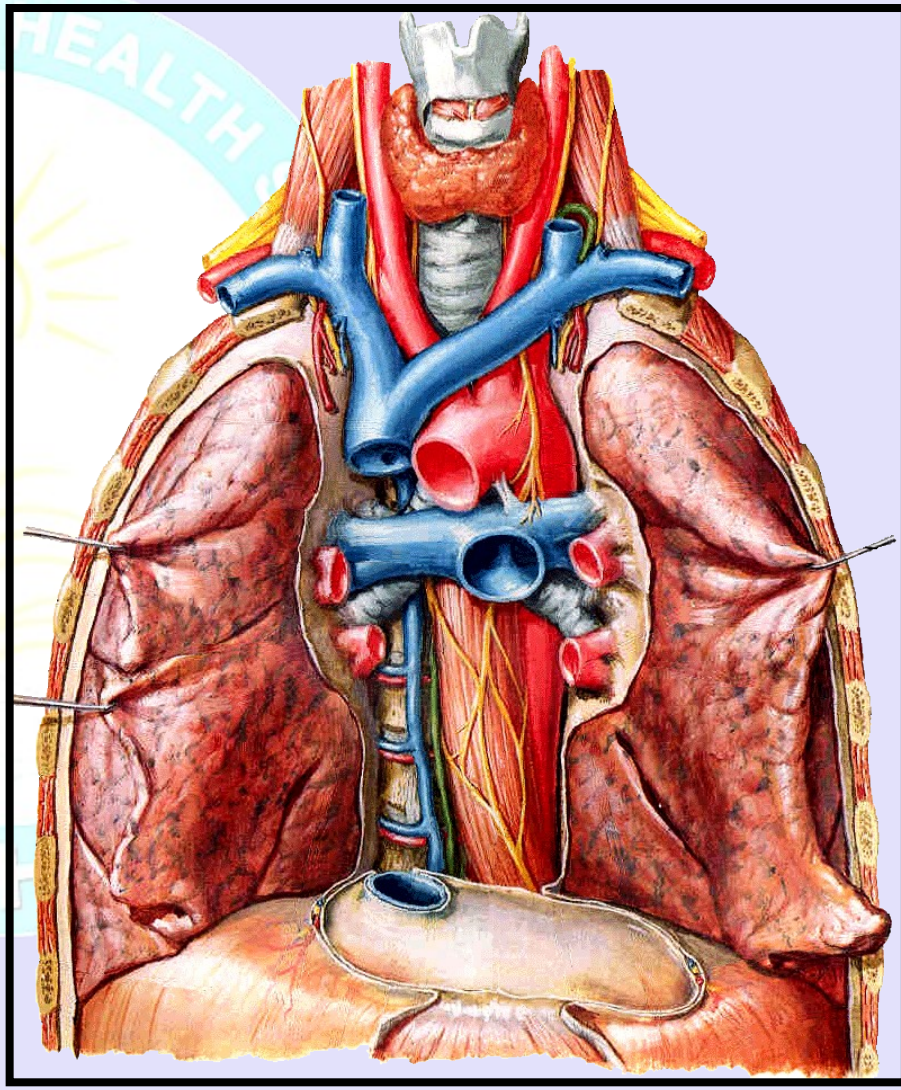
- Lies within posterior mediastinum.
- Begins at level of sternal angle.
- Ends in front of thoracic vertebra 12
- Continuous with abdominal aorta.



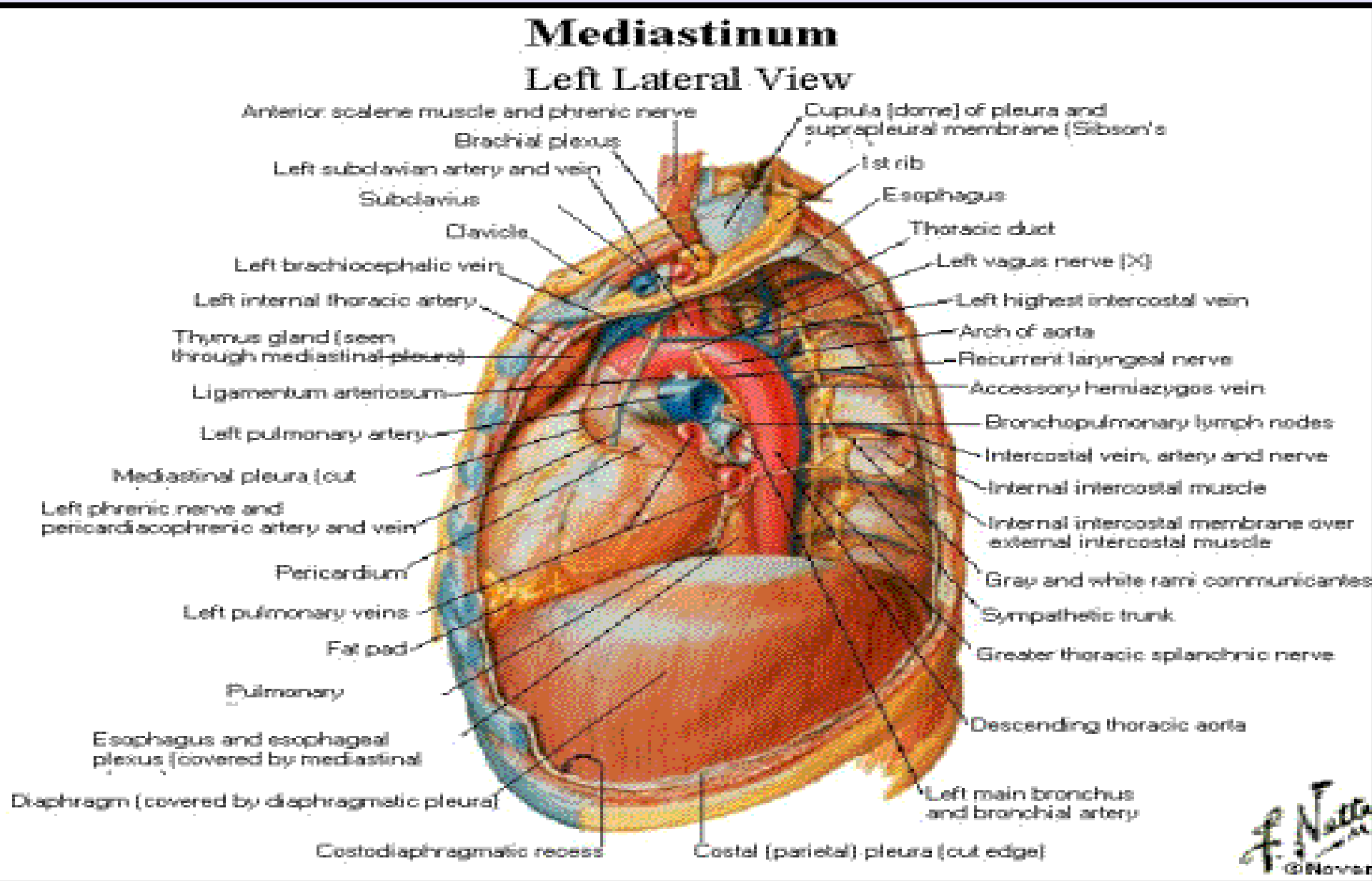
RELATIONS OF DESCENDING THORACIC AORTA

- **Anteriorly**
  - Left root of lung
  - Pericardium
  - Esophagus
- **Posterior**
  - Hemiazygos v.
  - Accessory hemiazygos v.
- **Right**
  - Azygos v.
  - Thoracic duct
- **Left**

  mediastinal pleura



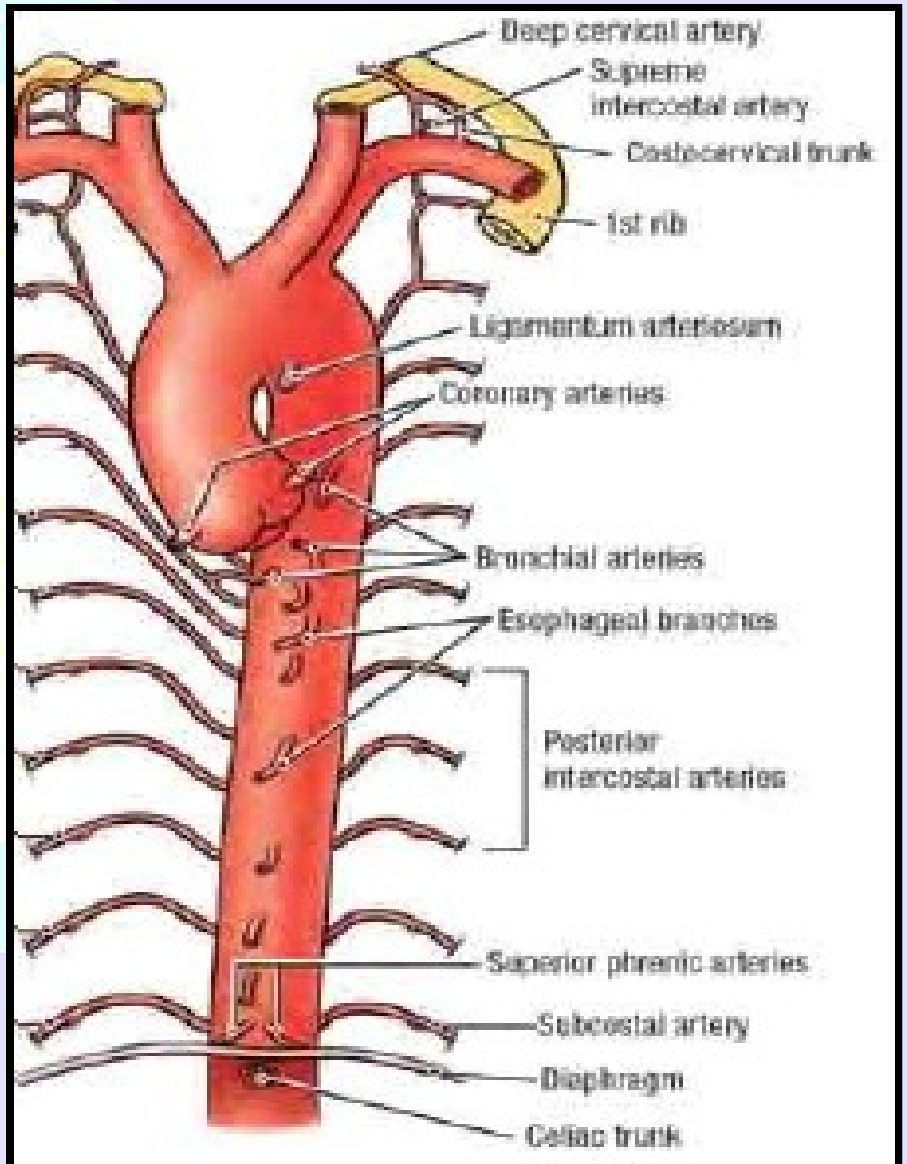
RELATIONS OF THORACIC AORTA



DESCENDING THORACIC AORTA

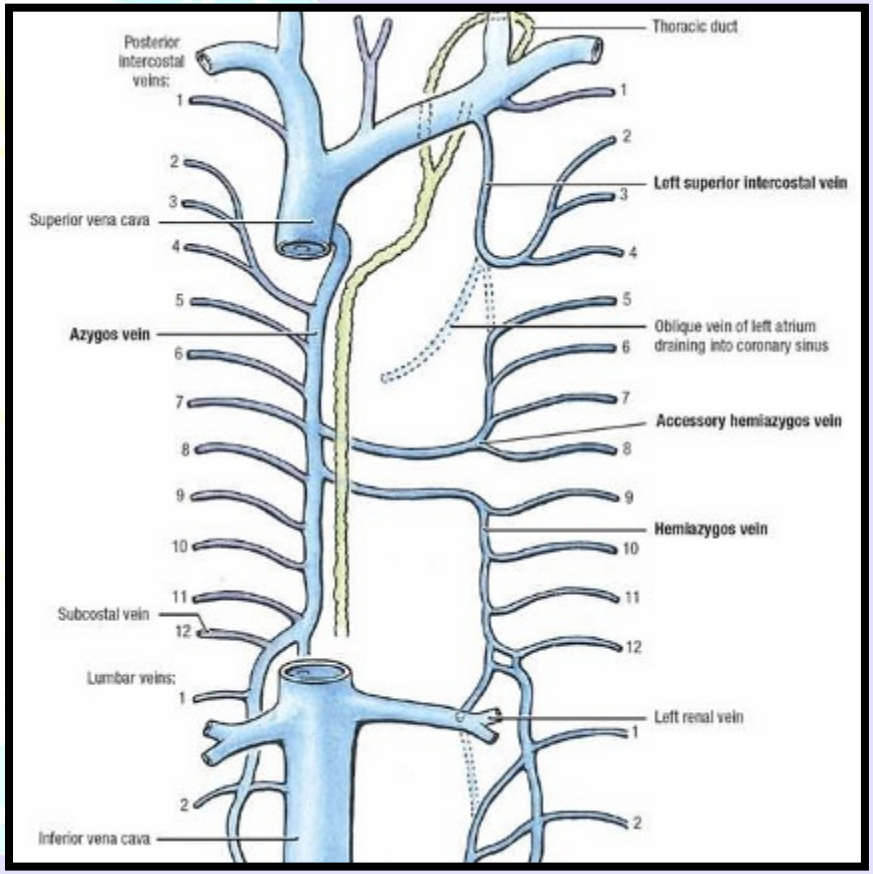
BRANCHES:

- **Paired intercostal arteries.**
- **Paired subcostal arteries.**
- **Two or more bronchial arteries.**
- **Two to five esophageal arteries.**



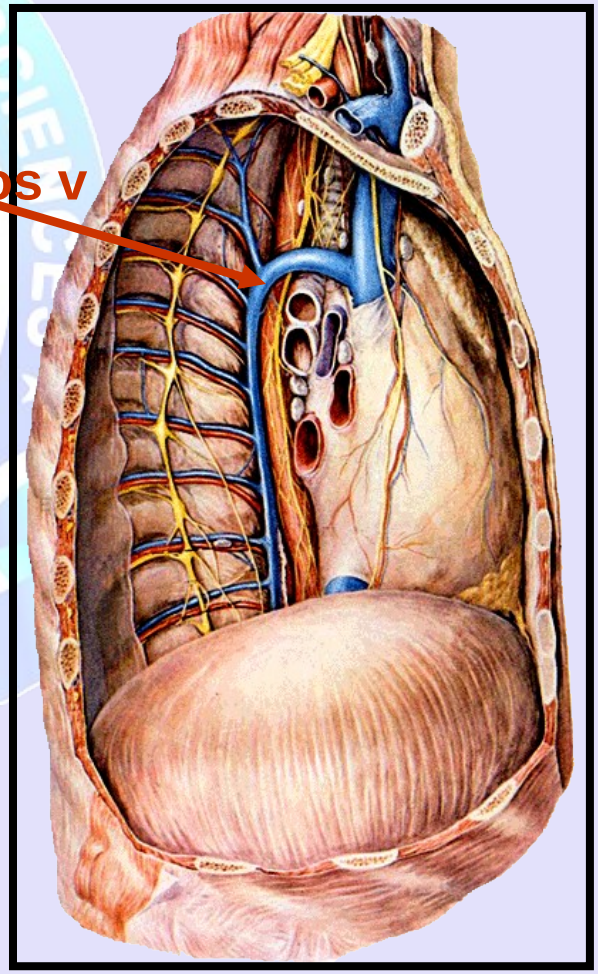
Azygos system of veins

- The azygos system of veins consists of a series of longitudinal vessels on each side of the body that drain blood from the body wall and move it superiorly to empty into the superior vena cava.
- Blood from some of the thoracic viscera also enter into the azygos system, and there are anastomotic connections between azygos and other systemic abdominal veins



AZYGOUS VEIN

- Forms in abdomen
- From right subcostal and ascending lumbar veins.
- Drains all right posterior intercostal veins except first.
- Also receives blood from the bronchial and esophageal veins.



Azygos system of veins

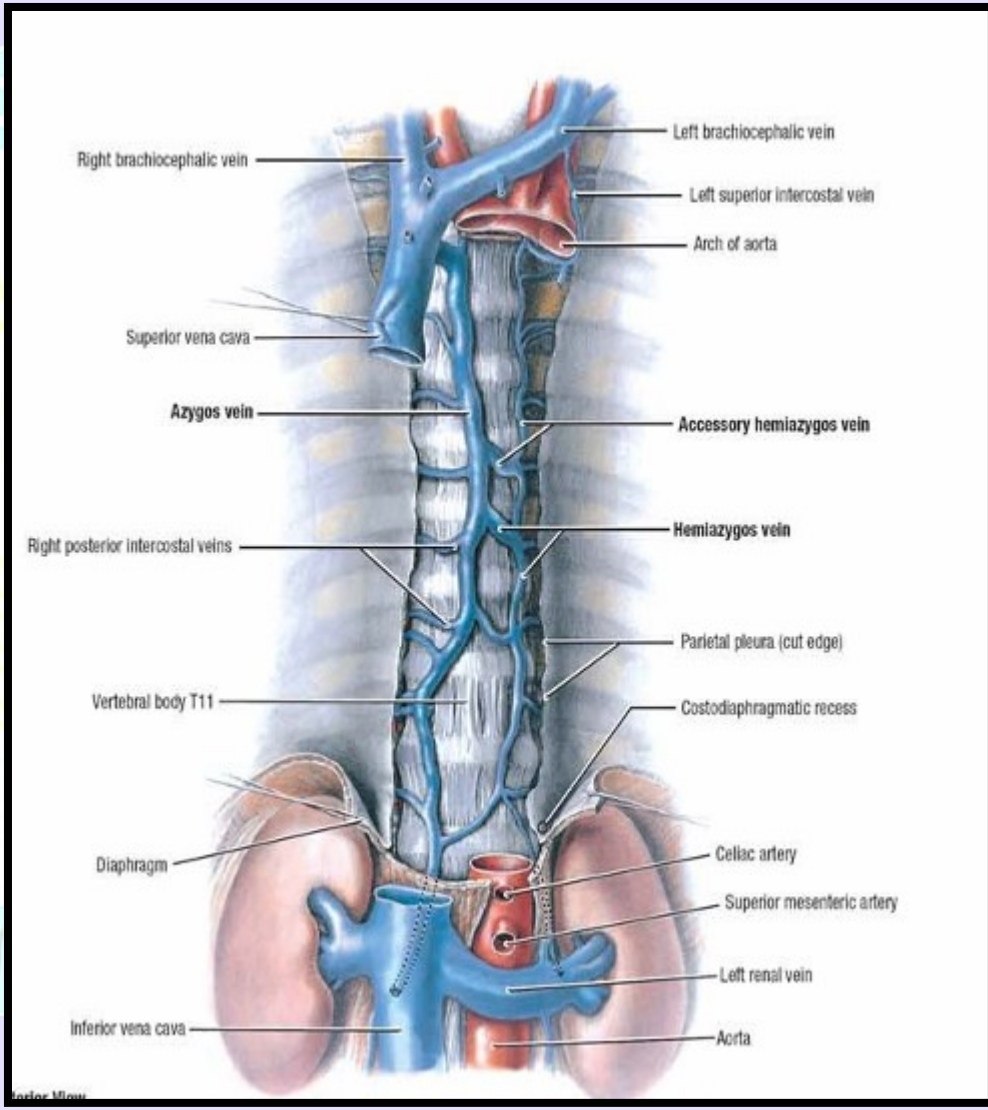
The azygos system of veins serves as an important anastomotic pathway capable of returning venous blood from the lower part of the body to the heart if the inferior vena cava is blocked. The major veins in the system are:

Azygos vein, on the right; and

Hemiazygos vein and

Accessory hemiazygos vein, on the left.

There is significant variation in the origin, course, tributaries, anastomoses, and termination of these vessels.



HEMIAZYGOUS VEIN

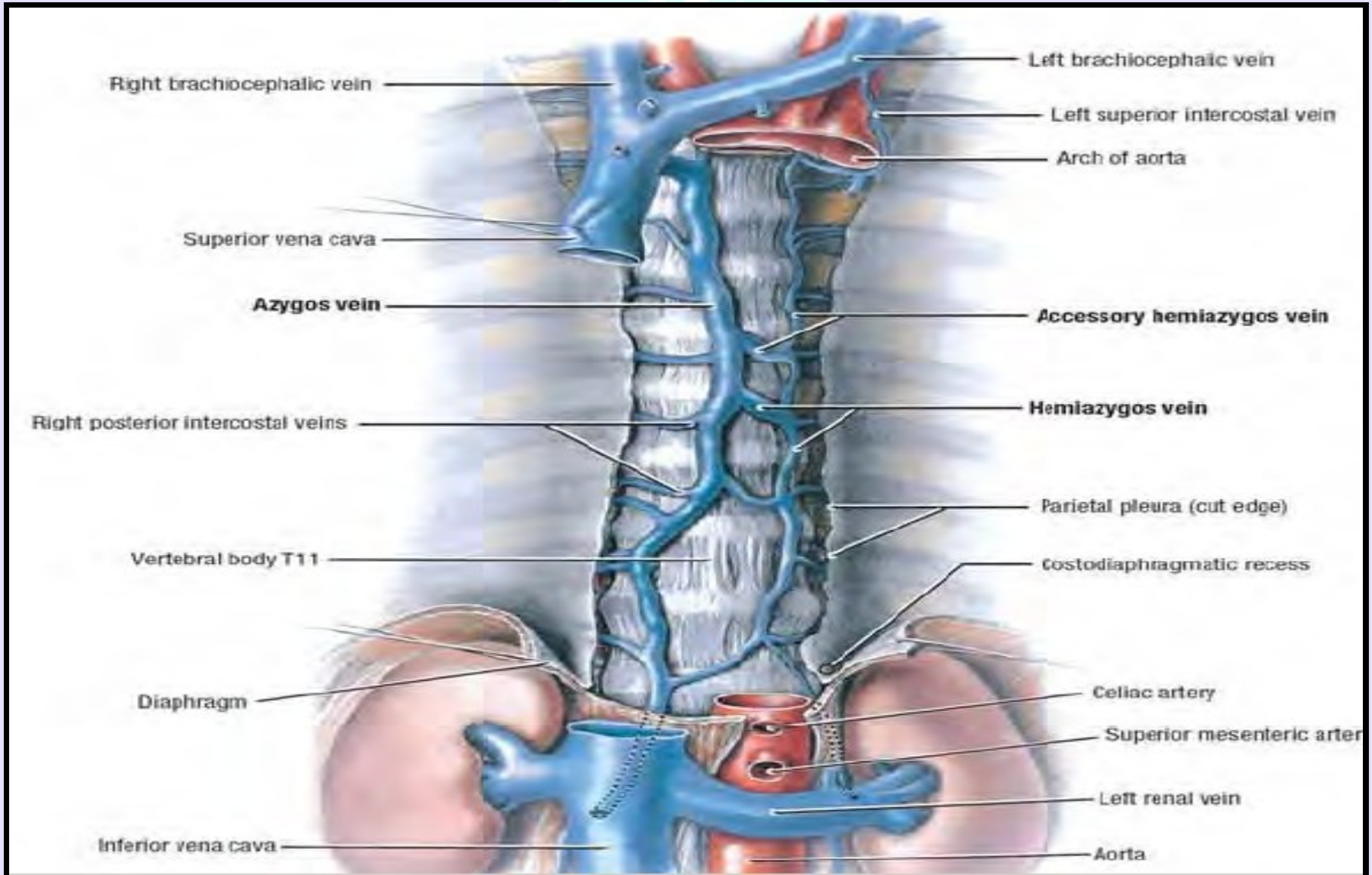
- Forms in abdomen
- From left subcostal and left ascending lumbar veins.
- Receives four posterior intercostal veins.
- Crosses over thoracic vertebrae at T8 level.
- Empties into azygos vein.

ACESSORY HEMIAZYGOUS VEINS

- Drains intercostal spaces 4-7(8) on left side.
- Crosses over thoracic vertebrae at level T7.
- Empties into azygos vein

- Note: Intercostal space 1 is drained by the supreme intercostal vein emptying into the brachiocephalic vein.

AZYGOUS, HEMIAZYGOUS AND ACESSORY HEMIAZYGOUS VEINS



LYMPH NODES OF POSTERIOR MEDIASTINUM

Posterior Mediastinal Lymph Nodes receives lymph from esophagus,

- **posterior aspect of the pericardium**
- **diaphragm and**
- **middle posterior ICS**

REFERENCES

- **CLINICAL ORIENTED ANATOMY BY K.L.M**

6TH EDITION

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THANK YOU